



# Geochemical and health risk assessment of fluoride and nitrate toxicity in semi-arid region of Anantapur District, South India

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## ABSTRACT

This paper reports to the fluoride and nitrate concentration in groundwater resources of the semi-arid region of Anantapur district, South India, and its related potential health risk assessment issues with the resident peoples. For this purpose, 50 groundwater samples were analyzed for hydrochemical parameters including major ions, fluoride and nitrate. The results revealed that most of the groundwater samples are alkaline in nature. Hydrochemical type of groundwater in the study area mainly  $\text{Na}^+ - \text{K}^+ - \text{Cl}^- - \text{SO}_4^{2-}$  and few samples belong to  $\text{Na}^+ - \text{K}^+ - \text{HCO}_3^-$ . Non carcinogenic health risks due to fluoride and nitrate exposure through consumption of groundwater were assessed using the USEPA method. In the study area groundwater samples, fluoride and nitrate concentration ranges from 1.2 to 5.9 mg/L and 36 to 92 mg/L. Based on these results, 86% of fluoride ( $>1.5$  mg/L) and 94% of the nitrate ( $>45$  mg/L) in groundwater samples exceed the permissible limit. The Total Hazard Index (THI) quotients for adults, children and infants had health hazards ( $\text{THI} > 1$ ) in 98%, 88%, and 80% of samples, respectively. Hence, proper precautionary measures have to be taken to control health risk in this area.

## 1. Introduction

Groundwater is used for drinking and irrigation purposes in arid and semi-arid regions are increasing considerably over around the world [22]. It evaluated that around 1.5 billion people rely on groundwater for irrigation and drinking purpose all over the world. In India groundwater constitutes nearly 80% for domestic purposes, around 45% of all agricultural water irrigating 39 million hectares [21,36,67]. Urban and rural areas rely on groundwater for drinking and around 90% of the rural people depend on groundwater for domestic purposes [57]. Due to overuse, fresh water is getting contaminated has become a scanty commodity and continued irrigation practices may also lead to groundwater pollution [41].

Groundwater pollution is associated with many sources like geogenic or anthropogenic source, nitrogen pollutants, agricultural pesticides, dissolution or weathering of the rocks and soil [4,5,14,18,42,64,65]. The quality of groundwater is of major concern in choosing the applicability of groundwater for a particular purpose and water for drinking purpose must be devoid of harmful elements, excessive minerals, living and non living organisms which may impact human health [27]. Contamination due to

toxic metals and metalloids is of huge health problem noticed in semi-arid area of Anantapur district, Andhra Pradesh [30,58–60]. Owing to natural and anthropogenic activities like agricultural and industrial activities, groundwater quality is slowly deteriorating, leading to water shortage crisis and numerous environmental health implications [24,47,70,71]. As public health directly linked with the quality of drinking water, its systematic monitoring is very essential.

Globally, since two decades, groundwater pollution by fluoride and nitrate is of major public health concern. In view of its importance, these studies have been carried out in arid and semi-arid areas around the world [12,17,19,25,37–39,63]. Fluoride concentration rely on various factors such as availability of circulating water to fluoride containing minerals, rock-water interaction mechanism, evaporation and evapotranspiration, complexation of fluoride ion with aluminum, beryllium, ferric iron and a series of mixed fluoride hydroxide complex with Boron [15]. Fluorides occur in three forms, namely fluorspar or calcium fluoride ( $\text{CaF}_2$ ), apatite or rock phosphate [ $\text{Ca}_3\text{F}(\text{PO}_4)_3$ ] and cryolite ( $\text{Na}_3\text{AlF}_6$ ) [57]. Though fluoride is required for normal mineralization of bones and formation of dental enamel in little quantities, when consumed in higher dosages ( $>1.5$  mg/L), it causes

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